Table 5.13 THEOPHYLLINE: - probabilities of non-acceptable measurement values (Acceptance interval ± 25%).

	Concentration (µg/l)					
Parameters	**0.05**	**0.10**	**0.50**	**1.00**	**2.50**	**10.00**
Lower quantity	1.8110	2.8070	4.1292	2.9164	2.3335	5.6003
Higher quantity	0.3280	1.0382	3.0106	2.8854	2.2141	4.2173
Interval std. dev.	0.0117	0.0130	0.0350	0.0862	0.2749	0.5093
Number of *df*	7.01	9.59	7.02	5.69	10.91	9.22
Below lower bounds	5.65%	0.97%	0.22%	1.44%	1.99%	0.02%
Above upper bounds	37.63%	16.23%	0.98%	1.49%	2.45%	0.11%
Global risk	43.28%	17.20%	1.20%	2.93%	4.45%	0.12%

measurements, the simplest method consists in adding the following lines and formulas to the end of the Resource H worksheet as shown in Resource J. It is applied to the theophylline level of 0.1 µg/l as the rest of the program. 0.97% of measurements that are below the limit of 0.075 µg/l, i.e. the lower limit of acceptance, and 16.23% exceed 0.125 µg/l. These results can be interpreted as an assessment of the risk taken by the analyst to produce results that would not meet the requirements of the end-user, bearing in mind that this is a probability and not a certainty!

Resource J Probability of nonacceptable measurements (Excel).

	A	B	C	D	E	F
47	**Resource J: Probability of non acceptable results**					
48	Assigned Reference Value (µg/L)	0.100	=B2			
49	Recovered concentration	0.1115	=B23			
50	b-ETI standard deviation (sTI)	0.013003	=B39			
51	Number of degrees of freedom (NE)	9.59	=B35			
52	Acceptance proportion	25%				
53	Acceptance lower bound	0.075	=B48*(1-B52)			
54	Acceptance upper bound	0.125	=B48*(1+B52)			
55	Lower quantile	2.807	=(B49-B53)/B50			
56	Upper quantile	1.038	=(B54-B49)/B50			
57	Probability Student L	1.0%	=TDIST(B$55;ROUNDDOWN($B$51;0);1)			
58	Probability Student U	0.9%	=TDIST(B$55;ROUNDUP($B$51;0);1)			
59	Percentage of too low values	0.97%	=B$57-((B$57-B$58)*($B$51-ROUNDDOWN($B$51;0)))			
60	Probability Student L	16.3%	=TDIST(B$56;ROUNDDOWN($B$51;0);1)			
61	Probability Student U	16.2%	=TDIST(B$56;ROUNDUP($B$51;0);1)			
62	Percentage of too high values	16.23%	=B$60-((B$60-B$61)*($B$51-ROUNDDOWN($B$51;0)))			

To conclude, the question of the choice of the type of tolerance interval.

– To conduct the validation of a method, we recommend using the β-ETI because it allows to highlight a set of parameters, such as the number of effective measurements or the variance ratio, which are especially useful at optimizing the experimental design. A method for estimating MU can also be easily derived from this type of TI by modifying the probability $\beta\%$ as explained in Section 7.2.

- The advantage of the β-γ-CTI is that it includes a confidence level, which is useful for knowing the bounds of the entire population of measurements for the same sample. This is the situation encountered with control charts where the same reference material is going to be analyzed several times in a row in the context of quality control. With a coverage probability of $\beta\% = 80\%$ and a confidence level $\gamma\% = 95\%$, it is possible to define, for a given concentration level, the bounds of an interval where, for 95% of 4 QC out of 5 would lie. According to FDA recommendations, it is even acceptable to take $\beta\% = 67\%$ and $\gamma\% = 95\%$ so that 2 QC out of 3 will fall within the acceptance interval [6]. But the algebraic form of β-γ-CTI does not allow to define a strategy for optimizing the experimental design.

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6

Measurement Uncertainty (MU)

6.1 Principle of Measurement Uncertainty

Metrology is "the science of measurement." Its modern history goes back to 1875, with the signature of the Metre Convention and the creation of the Bureau International des Poids et Mesures (BIPM). The Metre Convention is an international treaty, currently signed by about 64 countries which establishes the BIPM as the world authority for action in the field of metrology. The practical role of BIPM is to develop and control international measurement standards. They are required to cover an ever-expanding range of applications and to provide proof of equivalence between the national standards of different countries.

BIPM is also in charge of developing and managing the International System (SI) of units. Unfortunately for analysts, it was not until 1971 that, during the 14th BIPM meeting, it was finally decided to address the chemical measurement issue and introduce a new unit, the mole, about 100 years after other existing units. This delay illustrates the current challenges of chemical metrology.

In 1977, several scientists pointed out in a note that there was no consensus among measurers on the expression of the *error of measurement*. It was then asked by the BIPM to address this problem with other national metrology institutes and to issue a recommendation. A detailed questionnaire was prepared covering the topics involved and circulated to 32 national metrology institutes recognized as having an interest in the subject, supplemented by five international organizations.

The importance of having an internationally accepted procedure for expressing a new concept called measurement uncertainty (MU) was recognized by all participants. It then remained to construct the appropriate method to achieve this objective. A working group was created dedicated to this purpose and it prepared a recommendation that led to the *Guide to the expression of uncertainty in measurement*, abbreviated GUM, published in 1995. The emergence of MU is thus the result of a conceptual evolution that spans almost two decades. The current version of the GUM dates from 2020 but is currently under revision [1].

Before MU was considered relevant to analytical sciences, the concept of total analytical error (*TAE*) had been introduced around 1974 [2]. The reasoning behind this parameter is rather classical, as it is a perfect illustration of what is called the "error approach" in GUM.

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